PROGRAM: B.Tech
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CASE STUDY

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Doppler Effect in Analog Communication Systems

Introduction:

The Doppler Effect is an important phenomenon in communication systems that occurs when there is relative motion between the transmitter and receiver. Due to this motion, the frequency observed at the receiver is different from the transmitted frequency. This effect becomes very significant in systems such as mobile communication, satellite communication, radar systems, and aircraft communication.

In analog communication systems, Doppler shift mainly affects frequency-modulated signals and can cause distortion, frequency offset, and loss of signal quality if not properly compensated.

Theory of Doppler Effect

When a transmitter and receiver are moving relative to each other, the received frequency is given by the Doppler shift equation:

$f_d = (v/c)f_c$

Where:

- f_d = Doppler frequency shift
- v = relative velocity between transmitter and receiver
- c = speed of light
- f_c = carrier frequency

The received frequency is given by:

$f_r = f_c \pm f_d$

If the transmitter and receiver move towards each other, the received frequency increases. If they move away from each other, the received frequency decreases.

Block Diagram:

Message Signal
↓
Modulator
↓
RF Transmitter
↓
Channel with Noise + Doppler Shift
↓
RF Receiver
↓
Frequency Correction (AFC / PLL)
↓
Demodulator
↓
Output Message Signal

Effect of Doppler in Analog Communication Systems

- **Effect on Amplitude Modulation (AM)**

(Amplitude Modulation) systems are less affected by Doppler shift because information is carried in amplitude, not frequency. However, Doppler shift can still cause tuning problems in receivers.

- **Effect on Frequency Modulation (FM)**

(Frequency Modulation) systems are highly affected because information is carried in frequency. Doppler shift changes the frequency and causes distortion in the received signal.

Communication System	**Effect of Doppler**
AM	Small
FM	Large
Mobile Communication	Moderate
Satellite Communication	Very High
Radar Systems	Used for speed measurement

Applications of Doppler Effect

The Doppler Effect is used in many real-life communication systems:

1. **Radar Systems :** Radar systems use Doppler shift to measure the speed of moving vehicles.
2. **Satellite Communication:** Satellites move at very high speed, so Doppler compensation is necessary.
3. **Aircraft Communication:** Aircraft communication systems experience Doppler shift due to high aircraft velocity.
4. **Mobile Communication:** When a person is moving in a car or train while receiving signals, Doppler shift occurs.

Problems Caused by Doppler Effect

1. Frequency Shift

- The transmitted frequency and received frequency become different.
- The receiver cannot tune properly to the signal.
- This causes **distortion, weak signal, and poor reception**.
- In FM, it causes improper demodulation.
- In AM, it causes incorrect signal detection.

2. Loss of Synchronization

- The receiver must stay synchronized with the transmitter frequency.
- Doppler shift changes the frequency continuously.
- This causes **synchronization errors**.
- Results in **noise, unclear audio, and signal loss**.

Methods to Reduce Doppler Effect

1. Phase Locked Loop (PLL)

- Phase Locked Loop is used to track the incoming signal frequency.
- It automatically adjusts the receiver frequency.
- Keeps the receiver locked to the transmitted signal.
- Used in **FM receivers, satellite communication, mobile communication**.

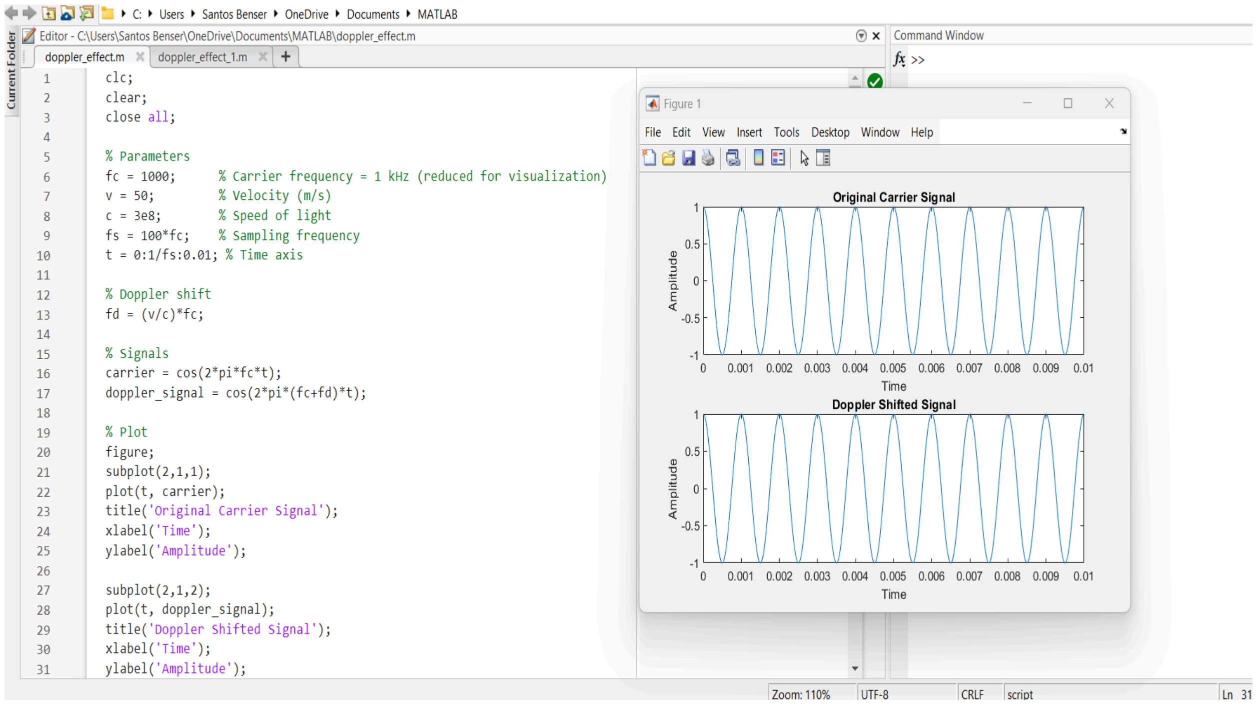
2. Automatic Frequency Control (AFC)

- Automatic Frequency Control automatically corrects frequency errors.
- It adjusts the local oscillator frequency in the receiver.
- Helps the receiver stay tuned to the correct frequency.
- Used in **radio receivers and analog communication systems**.

MATLAB Simulation of Doppler Effect

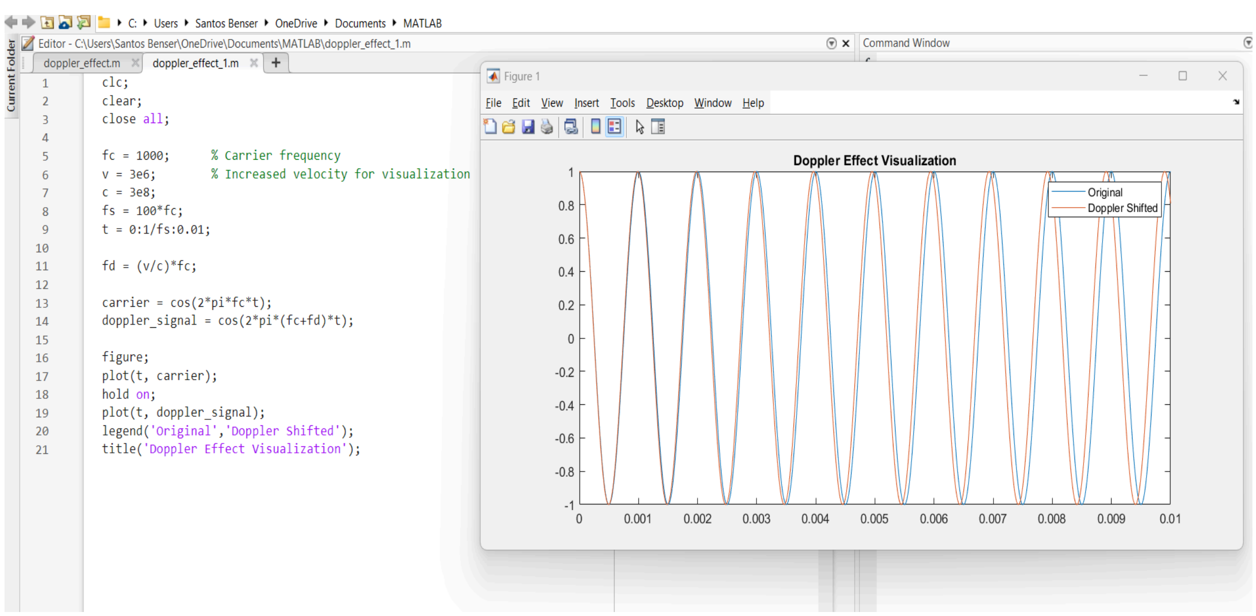
Case 1 → Car Example (Realistic Case)

- A car moves at around 20–50 m/s.
- When a car with a transmitter moves toward a receiver, the received frequency increases slightly.
- When it moves away, the frequency decreases slightly.
- But the change is very small, so it is not easily visible in the graph.



Code 2 → Rocket Example (High-Speed Case)

- A rocket or satellite moves at very high speed (thousands of m/s).
- So the Doppler shift becomes large.
- The received signal frequency changes significantly.
- This is clearly visible in the graph.



Result:

From the MATLAB simulation, it is observed that the Doppler Effect causes a change in the frequency of the received signal when there is relative motion between the transmitter and receiver. For low velocity (car), the Doppler shift is very small and the shifted signal is almost similar to the original signal. For high velocity (rocket/satellite), the Doppler shift is large and the shifted signal shows a higher frequency than the original signal. Thus, the Doppler shift increases with increase in velocity.

Conclusion:

The Doppler Effect is an important factor in analog communication systems, especially in mobile and satellite communication. It causes a shift in the received frequency due to relative motion between transmitter and receiver. This shift can cause distortion and synchronization problems in communication systems. However, techniques such as Phase Locked Loop (PLL) and Automatic Frequency Control (AFC) can be used to compensate for Doppler shift. MATLAB simulation shows the change in frequency due to Doppler Effect and helps in understanding its impact on communication systems.